Electronically Steerable Antenna Array using PCB-based MEMS Phase Shifters

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Abstract:

In this paper, MEMS phase shifters have been monolithically integrated with antennas on duroid substrate using printed circuit processing techniques. Specifically, 3-bit phase shifters have been integrated with a two-element folded slot antenna array for electronically scanned array (ESA) applications. As a proof-of-concept, a preliminary ESA prototype with beam steering of 4° at 10 GHz is presented. The proposed approach is suitable for fabrication of low cost, large area MEMS-based ESAs on Teflon or Polyimide like low dielectric constant printed circuit board (PCB) substrates.

1. Introduction

Electronically scanned/steerable antennas (ESA) or phased array antennas use electronic, micro-electromechanical, or material switches to alter the phase of individual radiating elements across an antenna aperture, and in so doing, enable the radiated beam to steer without any mechanical motion of the antenna system. ESAs have broad applicability for both commercial and military applications. Currently, ESAs are utilized in advanced military radars, cellular base stations, some satellite communications applications and other communication systems. Future uses include automotive anti-collision radar, smart navigation systems, and improved wireless communications. Multiple beams can be radiated by a single ESA to enable detecting, tracking, and communicating with multiple objects simultaneously. ESAs have been used for ground, ship, and airborne radars, and in several satellite systems. However, their complexity and cost has limited their wider use in space [1].

Phase shifters/switches are the devices in an ESA that enable the steering of an antenna beam without any mechanical motion of the antenna system. Phase shifters are critical components in ESAs and typically represent a significant amount of the cost of producing an antenna array. Phased array antennas or ESAs have traditionally used MMIC (Monolithic Microwave Integrated Circuit) phase shifters, which have two key weaknesses. They can contribute substantially to the cost of fabrication and tend to introduce a relatively high insertion loss, reducing the phased array antenna's Effective Isotropic Radiated Power.

In the past five years, RF Micro Electro Mechanical System (RF MEMS) switches and phase shifters that exhibit excellent characteristics including lower insertion loss and lower DC power consumption compared to their GaAs based electronic counterparts in previous ESAs have been demonstrated. Conventional RF MEMS phase shifters fabricated using surface micromachining techniques are not appropriate for integration with antennas on Printed Circuit Board (PCB) substrates, due to process compatibility, complexity and cost issues. Therefore, there is a need for low cost, printed-circuit compatible, RF MEMS technology for realization of low cost MEMS-based electronically scanned antenna array systems. Recently, MEMS phase shifters fabricated using printed circuit processing techniques have been reported by our group in [2]. These phase shifters are suitable for monolithic integration with low cost ESAs on Teflon or Polyimide like low dielectric constant PCB substrates.

In this work, an electronically steerable antenna array integrated with RF MEMS phase shifters fabricated using PCB processing techniques is presented.

2. MEMS Based ESAs

The complete configuration of the proposed PCB MEMS based ESA is shown in *Figure 1*. The ESA comprises of three components namely, the antenna element, the phase shifters and the power divider. In the following sections, the configuration of the ESA and design of various ESA components are described.

2.1. ESA Configuration

The ESA consists of three main layers: a Kapton E Polyimide film membrane (2 mils) with 3 μm thick copper metallization, an RT/Duroid 6002 Teflon substrate (30 mils) with 9 μm thick copper metallization, and a Polyflon adhesive spacer film (2 mils). The antenna structure and Coplanar Waveguide (CPW) lines are patterned using lithography techniques on the substrate copper metallization. The copper metallization on the bottom side of the Kapton membrane is patterned to define the MEMS varactor top electrode. The polyflon spacer film provides the initial gap height for the MEMS varactor. MEMS varactors are biased using 250 Å thick and 50 μm wide resistive Nichrome bias lines with a resistivity of 1.1×10^{-6} Ω-m. A 14 μm thick SU-8 dielectric layer is spin-coated on the CPW line metallization to provide a capacitive contact in the down state for the MEMS varactors. These three layers, namely Kapton film, Polyflon film, and Duroid, are laminated by thermo-compression bonding to obtain the ESA prototype. Details of the fabrication and assembly procedures for similar structures are described in [3].

The phase shifter has been implemented using MEMS varactors fabricated using printed circuit based MEMS technology [2].

In this configuration, a 3-bit has been designed by cascading three one-bit phase shifter units. A one-bit phase shifter unit consists of two identical MEMS varactors separated by a center-to-center distance of quarter wavelength as indicated in *Figure 1(a)*. The configuration in *Figure 1(b)* depicts the initial position of the MEMS varactor. For operation, a DC bias voltage is applied to pull down the MEMS varactor membrane such that the top electrode is in contact with the dielectric layer (down state). The capacitance increase in the down state provides the required phase shift in the phase shifter, which enables desired steering of the antenna beam.

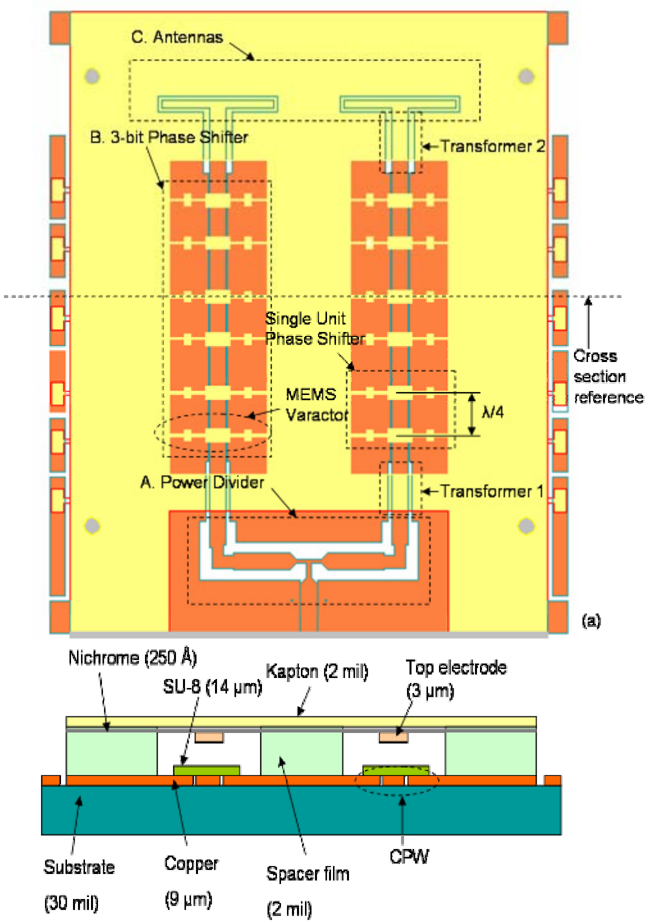

Figure 1. (a) Top view of ESA (b) Cross sectional view of ESA. (not to scale).

2.2. ESA Component Design

In this section, design and simulation of various ESA components such as power divider, phase shifters, and antenna elements are described.

A. Power Divider Design

A CPW power divider has been designed for feeding the two element antenna array. The power divider consists of a 50 Ohm input line and two 100 Ohm output lines. Each output line is matched to the 3-bit phase shifter using a quarter-wave transformer (Transformer 1 in *Figure 1*). To minimize the junction discontinuity effects, the configuration shown in *Figure 2* with narrow inductive sections at the junction of the power divider has been implemented. This configuration of the power divider minimizes the capacitive discontinuity effect at the T-junction (where the input 50 Ohm line branches into two 100 Ohm lines). Also, compensated right angled bends were used to connect the power divider and the phase shifters [4]. Air bridges (2 mil bond wires) are used to suppress the excitation of slot modes at the T-junction and right-angle bends in the power divider [4]. Electromagnetic simulation and design optimization of the power divider design have been carried out in ADS Momentum [6]. S-parameter results of the optimized power divider design (design parameters shown in *Figure 2*) are shown in *Figure 3*. The input return loss at port 1 and the insertion loss between the input port (port 1) and output ports (ports 2 and 3) are about 10 dB and 3.8 dB, respectively, at 10 GHz.

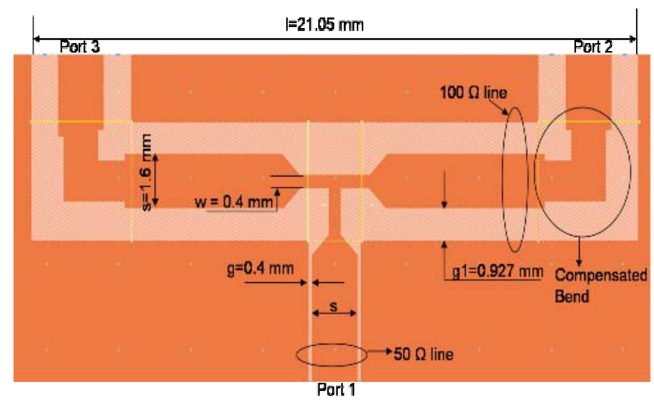

Figure 2. CPW power divider layout showing optimized design parameters.

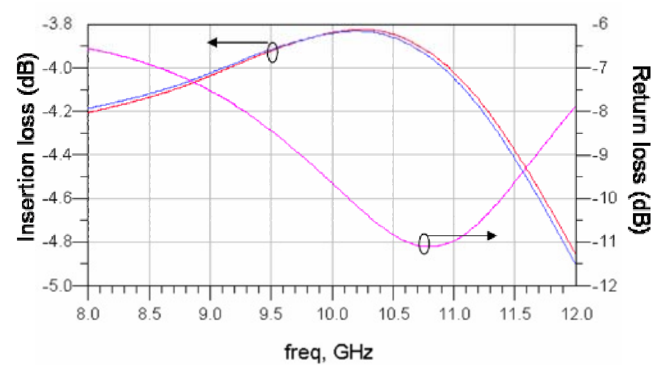

Figure 3. Return loss and Insertion loss for the Power Divider.

B. Phase Shifters:

The ESA consists of two 3-bit phase shifters connected to antenna elements. Each 3-bit phase shifter has been designed by cascading three one-bit phase shifter units. A one-bit phase shifter unit consists of two identical bi-stable MEMS varactors separated by a center-to-center distance of quarter wavelength as indicated in *Figure 1*.

In the bi-stable design, MEMS varactors provide two different capacitance values corresponding to the up position and down position of the MEMS varactor membrane. For operation, a DC bias voltage is applied to pull down the

MEMS varactor membrane such that the top electrode is in contact with the dielectric layer (down position). The phase shift depends on the difference between the two capacitance values provided by these two states.

Top view of the bi-stable MEMS varactor is shown in *Figure 4*. The dimensions of the CPW line are chosen to be S = 1.6 mm and G = 0.1 mm to provide characteristic impedance Zt = 50 Ω on RT/Duroid 6002 substrate (ε_r = 2.94). The electrode dimensions of the MEMS varactor are: l_1 = 1 mm and w_1 = 2 mm. Dimensions l_2 and w_2 are varied to obtain the required capacitance and are listed in *Table 1*.The flexure design dimensions are: L_1=1.375 mm, L_2= 0.75 mm, L_3 = 1.375 mm, L_4 = 1 mm and the flexure width w = 0.2 mm. The thickness of the SU-8 dielectric layer (ε_r=4.2) is 14 µm. The design parameters and capacitance values obtained for various MEMS varactor designs using ANSOFT Q3D [7] are shown *Table 1*. The phase shifts obtained at 10 GHz from ADS Momentum simulations [3] for phase shifters implemented using these MEMS varactors are also shown in *Table 1*. Insertion losses and phase delays obtained from ADS Momentum simulations are shown in *Figure 5* and *Figure 6*, respectively.

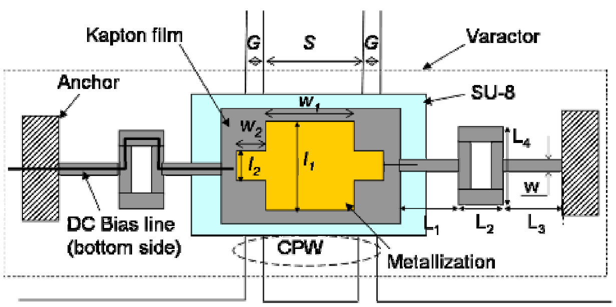

Figure 4. MEMS varactor top view (not to scale).

Table 1. Design parameters for the phase shifters.

Phase shifter index	Up State Capacitan-ce from (fF)	Down State Capacitance (fF)	w_2 (mm)	ℓ_2 (mm)	Phase shift @10 GHz Ø(°)
1	70	167	0.1	0.1	16°
2	87	384	0.1	0.6	37°
3	119	784	0.15	1	81°

C. Antenna Element and array

The ESA consists of two half-wavelength, center-fed, folded slot antenna elements. Each antenna element is matched to the 3-bit phase shifter on a 50 Ω line using a quarter-wave transformer (Transformer 2 in Figure 1). A folded slot design was chosen over a simple center fed slot because of the impedance reduction obtained in the folded slot design when compared to a simple slot design [8]. The input impedance of a folded slot antenna is given by

$$Z_{in} = \frac{Z_{slot}}{N^2} \quad \ldots\ldots\ldots\ldots(1)$$

Where N is the number of slots, Z_{slot} is the impedance of a simple slot and Z_{in} is the input impedance of the folded slot structure. The two-element folded slot antenna array shown in *Figure 7* has been simulated in ADS Momentum [4] and the return loss is shown in *Figure 8*.

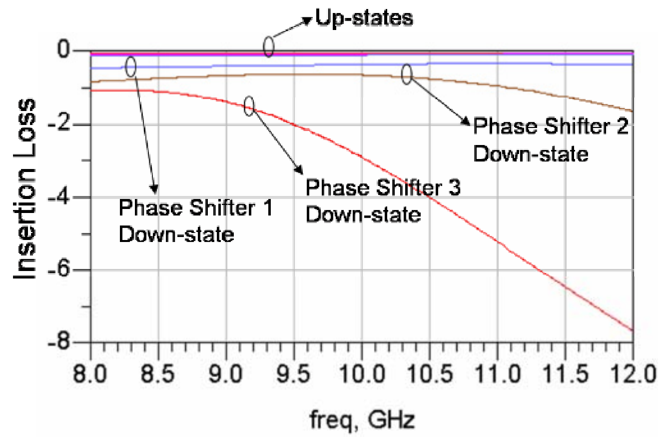

Figure 5. Insertion losses for various Bi-stable phase shifter units in the 3-bit phase shifter.

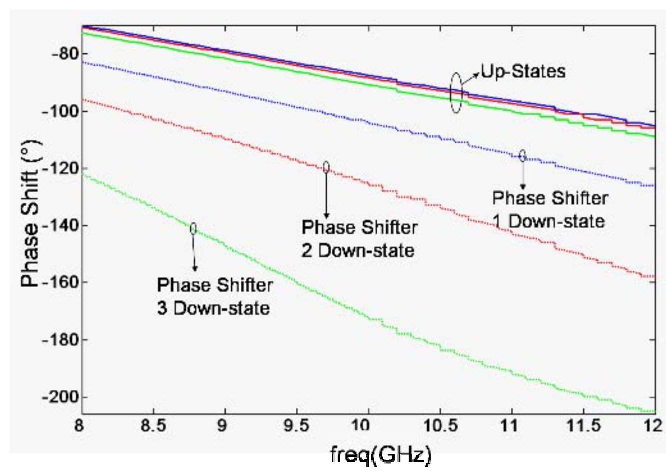

Figure 6. Phase delay for various Bi-stable phase shifter units in the 3-bit phase shifter.

As expected, identical return loss performance was observed for both antenna elements. The beam steering characteristics of the two-element antenna array for various phase shifts have been simulated using ADS [4]. In these simulations, the input signal to Port B is phase shifted by Ø with Port A fed by zero phase shifted signal as shown in *Figure 7*. The beam angle of the two element antenna array in the H-plane for various phase shifts are shown in *Figure 9*, where Ø is the phase shift of the signal fed at Port B in *Figure 7*. It can be observed that the beam angle shifts from 0° to 19.5° in the H-plane as the phase shift of the signal fed at Port B is varied from Ø =0° to 90°.

2.3. Results and Discussion

A photograph of the fabricated Electronically Steerable Antenna with overall dimensions is shown in

Figure 10. The measured return Loss and the H-plane radiation pattern of the ESA are shown in *Figure 11* and *Figure 12*, respectively. The Return loss measurements have been carried out using an HP 8510C network analyzer.

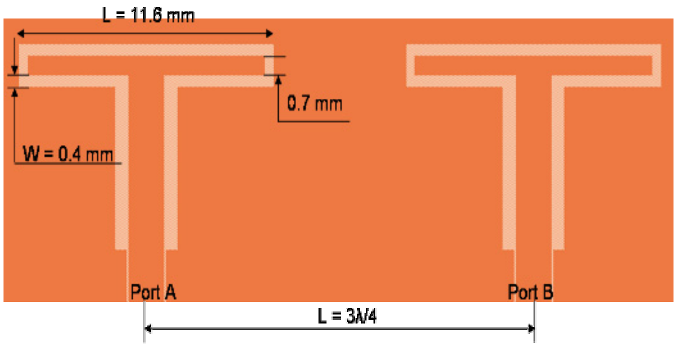

Figure 7. Two-element folded slot antenna array with quarter-wave transformer.

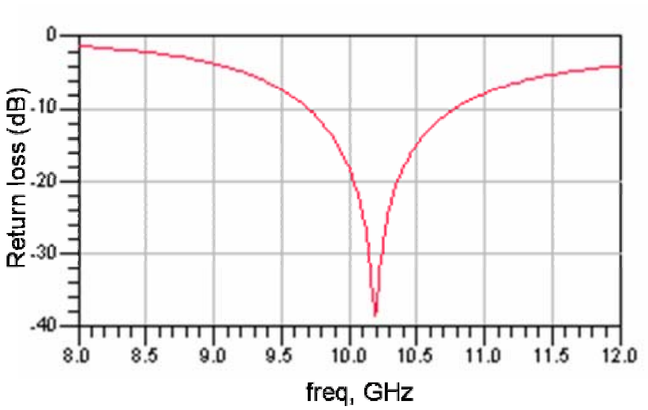

Figure 8. Return loss for a single folded slot antenna in the two-element array.

It can be observed from *Figure 10* that the return loss of the ESA is 17 dB at the resonant frequency of 10.5 GHz. The antenna radiation patterns in two different states, shown in *Figure 12*, have been measured using Auburn University's pattern measurement system located in a 12m x 5m x 2.5m anechoic chamber. The *Beam 1* and *Beam 2* patterns correspond to zero phase shifted signal and phase shifted signal, respectively, fed to the right antenna element in the two-element array. It can be seen from *Figure 12* that the antenna beam shifts by 4° when the MEMS varactors in phase shifter units 2 and 3 (shown in *Figure 10*) are in the down state. The DC voltages required to pull down the MEMS varactors in the phase shifter units 2 and 3 are in the range of 160-170 V respectively. The estimated phase shift is 16° corresponding to the measured beam shift of 4°. The measured beam shift is lower than the expected value of 16° corresponding to a phase shift of 75°. The low phase shift value can be attributed to low capacitance ratio resulting from the surface roughness of the duroid substrate. However, the preliminary prototype presented here demonstrates monolithic integration of MEMS phase shifters with antennas on duroid substrate. Currently, work is in progress to improve the current design and the results will be presented at the conference.

3. Conclusion

In this paper, monolithic integration of MEMS phase shifters with antennas on duroid substrate has been demonstrated for ESA applications. A preliminary ESA prototype with beam steering of 4° at 10.5 GHz has been discussed. The proposed design and configuration could be improved to provide beam steering angles of up to 20°.

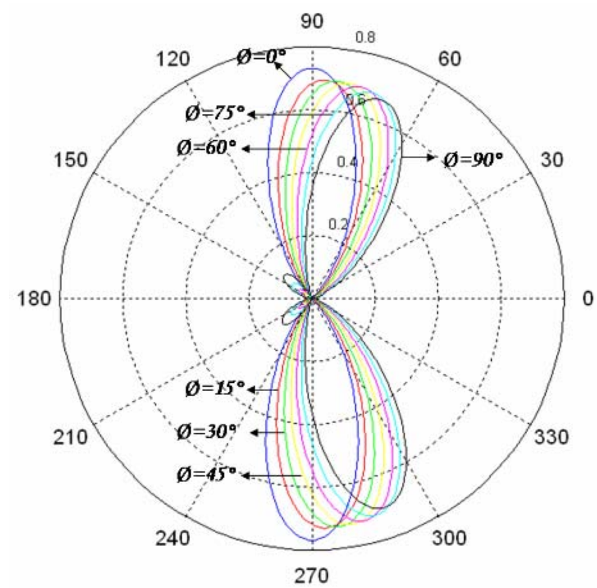

Figure 9.Simulated beam steering characteristics of the antenna array in the H-plane for various phase shifts (Port A fed with zero phase shifted signal and port Port B fed with the phase shifted signal (Figure 7))

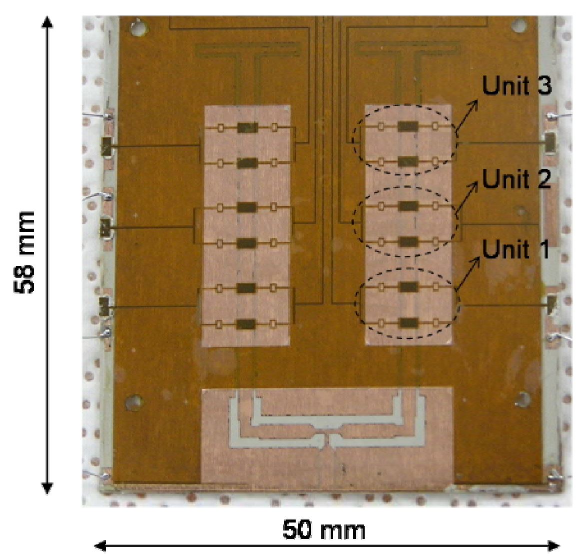

Figure 10. Fabricated Two-element X-band Electronically steerable Antenna Prototype.

Acknowledgements

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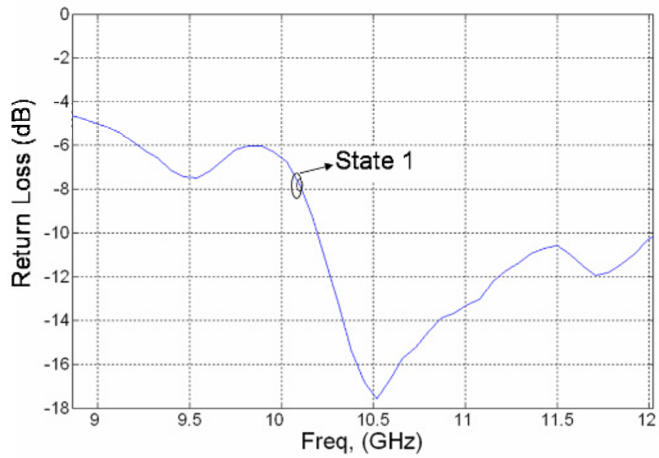

Figure 11. Measured return loss for the ESA prototype with all MEMS varactors in up position corresponding to zero phase shift condition.

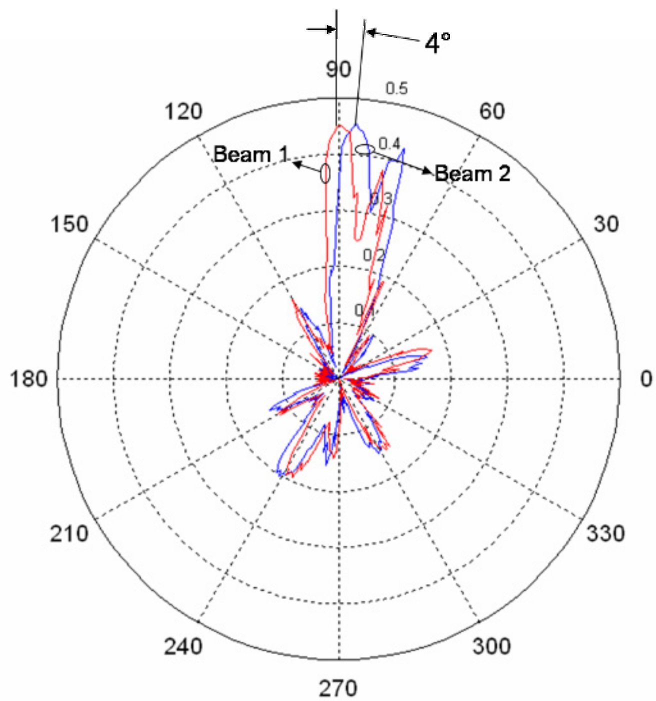

Figure 12. Antenna H-plane radiation patterns for the zero phase shifted signal (beam 1) and phase shifted signal (beam 2) fed to the right antenna element in the two-element array.

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